

Customer Shopping Details Analysis

(Python, SQL, Power BI)

Project Overview :

The objective of this project is to analyse customer shopping behaviour using a retail dataset containing 6,000 customer transactions. The analysis focuses on identifying purchasing patterns, customer preferences, sales trends, subscription behaviour, seasonal demand, payment preferences, and business opportunities.

The project uses **MySQL** for data storage and SQL-based analysis, while **Power BI** is used to build interactive dashboards that help stakeholders make data-driven business decisions.

Tools & Technologies

Tool	Purpose
MySQL Workbench	Database Management
SQL	Data Cleaning & Analysis
Power BI	Dashboard & Visualization
Python	Dataset Preparation

Dataset Summary:

Dataset Name: Customer Shopping Dataset

Total Records : 6000

Total Columns : 18

- Customer ID
- Age
- Gender
- Item Purchased
- Category
- Purchase Amount (Rs.)
- Location
- Size
- Color
- Season
- Review Rating
- Subscription Status
- Shipping Type
- Discount Applied
- Promo Code Used
- Previous Purchases
- Payment Method
- Frequency of Purchases
-

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Exploratory Data Analysis using Python :

We began with data preparation and cleaning in Python:

Data Loading: Imported the dataset using pandas.

Initial Exploration: Used `df.info()` to check structure and `.describe()` for summary statistics.

```
PS C:\Users\Kaushal\Desktop\kaushal\data analytis\project\customer_shopping_Details> python main.py
```

	Customer ID	Age	Purchase Amount (Rs.)	Review Rating	Previous Purchases
count	6000.000000	6000.000000	6000.000000	6000.000000	6000.000000
mean	103000.500000	44.01250	7700.60250	2.99325	24.812833
std	1732.195139	15.34077	4196.86537	1.15753	14.787740
min	100001.000000	18.00000	503.00000	1.00000	0.000000
25%	101500.750000	31.00000	4028.00000	2.00000	12.000000
50%	103000.500000	44.00000	7678.00000	3.00000	25.000000
75%	104500.250000	57.00000	11347.25000	4.00000	38.000000
max	106000.000000	70.00000	14995.00000	5.00000	50.000000

Missing Data Handling:

Checked for null values and imputed missing values in the Review Rating column using the median rating of each product category.

Column Standardization:

Renamed columns to **snake case** for better readability and documentation.

Feature Engineering:

- Created **age_group** column by binning customer ages.
- Created **purchase_frequency_days** column from purchase data.

Data Consistency Check:

Verified if discount applied and promo_code_used were redundant; dropped promo_code_used.

Database Integration:

Connected Python script to PostgreSQL and loaded the cleaned Data frame into the database for SQL analysis.

Data Analysis using SQL (Business Transactions) :

We performed structured analysis in PostgreSQL to answer key business questions:

revenue generated by male vs female customer :

Result Grid		Filter Rows:
gender	revenue	
Male	23215414	
Female	22988201	

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customer used a discount but still spend more then the average purchase amount :

	customer_id	purchase_amount
▶	100002	11036
	100005	10594
	100006	14965
	100008	13500
	100010	11844
	100017	10980
	100018	7909
	100019	11598

top 5 product with the highest review rating :

	item_purchased	average product rating
▶	Hoodie	3.08
	Kurti	3.02
	Jeans	3
	T-Shirt	2.97
	Laptop Bag	2.97

average purchase amount between standard and express shipping :

	shipping_type	avg_purchase_amt
▶	Express	7633.47
	Standard	7713.15

do subscribed customer spend more? compare average spend and total revenue between subscribed and non-subscribed:

	subscription_status	total_customer	average spend	total revanue
▶	No	3105	7771.41	24130240
	Yes	2895	7624.65	22073375

5 product have the highest percentage with the discount applied :

	item_purchased	discount rate
▶	Kurti	49.48
	Laptop Bag	49.58
	Jeans	49.78
	Watch	51.98
	Sneakers	51.54

segment customer into new, returning and loyal based on their total number of previous purchase, and show the count of each segment:

	customer_sagement	number of customer
▶	Returning	1089
	Loyal	4779
	New	132

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3 most purchased product within each category:

Result Grid		Filter Rows:	Export:	
	item_rank	category	item_purchased	total_orders
	1	Accessories	Watch	883
	2	Accessories	Laptop Bag	835
	1	Clothing	Jeans	900
	2	Clothing	T-Shirt	873
	3	Clothing	Kurti	865
	1	Footwear	Sneakers	811

customer who are repeat buyers(more than 5 previous purchases) also like to subscribe:

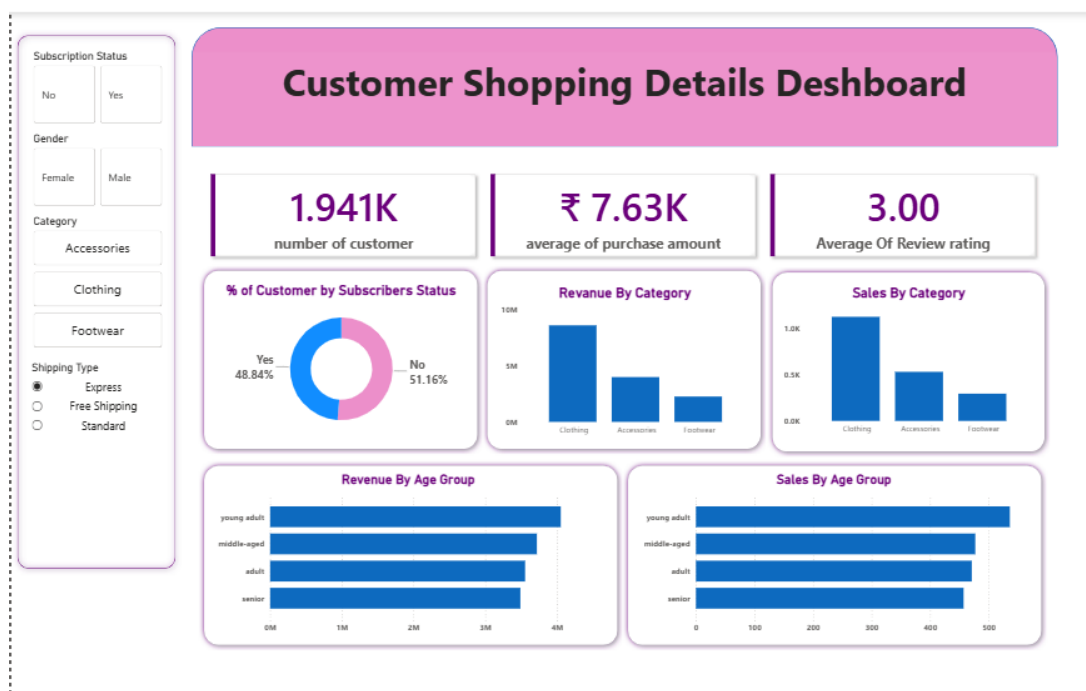
	subscription_status	repeat_byuers
▶	No	2764
	Yes	2613

revenue contribution of each age group :

	age group	total rewanue
▶	adult	11146253
	senior	11473024
	young adult	12312738
	middle-aged	11271600

Dashboard in Power BI :

Finally, we built an interactive dashboard in **Power BI** to present insights visually.



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Business Recommendations :

- **Boost Subscriptions** – Promote exclusive benefits for subscribers.
- **Customer Loyalty Programs** – Reward repeat buyers to move them into the “Loyal” segment.
- **Review Discount Policy** – Balance sales boosts with margin control.
- **Product Positioning** – Highlight top-rated and best-selling products in campaigns.
- **Targeted Marketing** – Focus efforts on high-revenue age groups and express-shipping users.